```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df=pd.read_csv('/content/supply_chain.csv')
df.head(2)
```

	Product type	SKU	Price	Availability	Number of products sold	Revenue generated	Customer demographics	Stock levels	Lead times	Order quantities	...	Location	Lea tim
0	haircare	SKU0	69.808006	55	802	8661.996792	Non-binary	58	7	96	...	Mumbai	2
1	skincare	SKU1	14.843523	95	736	7460.900065	Female	53	30	37	...	Mumbai	2

2 rows × 24 columns

```
!pip install plotly
```

```
Requirement already satisfied: plotly in /usr/local/lib/python3.13/dist-packages (5.24.1)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.13/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from plotly) (26.3)
```

```
import plotly.express as px
import plotly.io as pio
import plotly.graph_objects as go
pio.templates.default='plotly_white'
```

```
df.describe()
```

	Price	Availability	Number of products sold	Revenue generated	Stock levels	Lead times	Order quantities	Shipping times	Shipping costs	Lead time
count	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000
mean	49.462461	48.400000	460.990000	5776.048187	47.770000	15.960000	49.220000	5.750000	5.548149	17.080000
std	31.168193	30.743317	303.780074	2732.841744	31.369372	8.785801	26.784429	2.724283	2.651376	8.846251
min	1.699976	1.000000	8.000000	1061.618523	0.000000	1.000000	1.000000	1.000000	1.013487	1.000000
25%	19.597823	22.750000	184.250000	2812.847151	16.750000	8.000000	26.000000	3.750000	3.540248	10.000000
50%	51.239831	43.500000	392.500000	6006.352023	47.500000	17.000000	52.000000	6.000000	5.320534	18.000000
75%	77.198228	75.000000	704.250000	8253.976921	73.000000	24.000000	71.250000	8.000000	7.601695	25.000000
max	99.171329	100.000000	996.000000	9866.465458	100.000000	30.000000	96.000000	10.000000	9.929816	30.000000

```
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```

now let's get started with analysis the supply chain by lokinng at the relationship b/w the price of the products and the revenue genreted by them:

```
#!pip install statsmodels
```

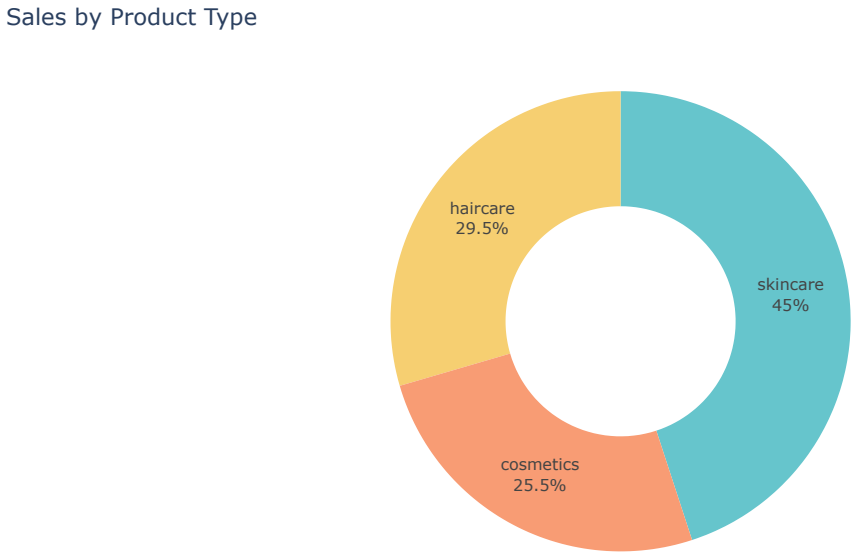
```
fig=px.scatter(df,x='Price',
               y='Revenue generated',
               color='Product type',
               hover_data=['Number of products sold'],
               trendline='ols')
fig.show()
```



thus,the company derives more revenue from skincare products,and the higher the price of skincare products,the more revenue they genrete.now let's have a look at the sales by product type: **bold text**

```
sales_data=df.groupby('Product type')['Number of products sold'].sum().reset_index()
#sales_data

pie_chart=px.pie(sales_data,names='Product type',
                 values='Number of products sold',
                 title='Sales by Product Type',
                 hover_data=['Number of products sold'],
                 hole=0.5,
                 color_discrete_sequence=px.colors.qualitative.Pastel)
pie_chart.update_traces(textposition='inside',textinfo='percent+label')
pie_chart.show()
```



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```
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```